Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/07/2024
PI Name:	Naithani, Sushma Ph.D.		
Project Title:	In Silico Modeling of Gene Networks and Mechanisms Associated With Plant Gravitropism		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	97333	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	04/06/2022	**End Date:**	04/05/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 04/05/2024 per NSSC information (Ed., 10/30/23).		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	80NSSC22K0891		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Gravity acts as an important determinant for shaping a plant's architecture. Understanding plant growth, development, and survival in response to altered/microgravity is crucial for developing sustainable bioregenerative life support systems for astronauts in long-term spaceflights, the International Space Station (ISS), and colonization of the Moon and Mars. The available research data suggest that plants can grow and complete their life cycles in the microgravity environment of spaceflights and the ISS; however, altered gravity and light conditions impact plant productivity, vigor, and starch storage organs (e.g., seeds and tubers). Furthermore, several independent studies on Arabidopsis (a few on rice and other species) mutants defective in hormone signaling and gravitropic response have contributed to the identification of the molecular components and mechanisms associated with plant gravitropism and how it is influenced by other tropism associated with environmental stimuli such as light, availability of water, and nutrients. In recent years, a few transcriptomic, proteomic, and metabolomic studies on plants subjected to altered gravity in spaceflights and on mutants defective in gravitropism have identified differential expression of genes associated with the fundamental biological processes of photosynthesis, phototropism, gas exchange, plant anchoring, pigment synthesis, and water and nutrient uptake. However, the lack of a system-level pathway network model for visualizing gravitropism (gravity-induced differential growth processes) and its connections with other tropisms hinders full utilization/analysis of the high-throughput transcriptomic and proteomic data. We propose to biocurate and build an in silico model of plant gravitropism response by analyzing the Omics datasets from Arabidopsis thaliana available at NASA GeneLab; https://) and based on a review of published literature. These pathways will be used for deriving the projection of pathways for an additional 106 species, ranging from unicellular photoautotrophs to higher plants, including several models and crop plants. The curated pathways and gene-orthology-based pathways will be hosted on the Plant Reactome knowledgebase (https://), an open-source pathway portal of the Gramene project. Plant Reactome can (i) serve as a nodal point for knowledge synthesis about gravitropism and associated genes, regulatory networks, and molecular mechanisms within the context of overall plant development and metabolism; (ii) provide a platform to space biology researchers in visualizing and analyzing Omics datasets; (iii) extend the knowledge of genes and pathways that influence gravity response or intersect with other regulatory mechanisms (phototropism, chemotropism, hydrotropism, and hormone signaling); and (iv) inform about evolutionary relevance of the conclusions derived from the Arabidopsis. Overall, our proposal will support space biology researchers to formulate data-driven hypotheses for designing plants suited for cultivation in controlled environments in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our project involves biocuration of pathways and gene-regulatory networks involved in plant gravitropism within the Plant Reactome Knowledgebase that serves as a free-of-cost, public resource for plant genomics scientists, breeders, and educators. In particular, in silico modeling of gene-gene networks and gravitropism associated pathways provide a conceptual framework for understanding how plant development and architecture is shaped by interaction of gravity with the plants overall genetic developmental program and other environmental stimuli. It is well aligned with NASA's space biology goals "Create new knowledge of how biological systems adapt to space that can translate into benefits for NASA Exploration Missions and for Earth applications through sponsoring competitively solicited Space Biology research" and "Train and inspire a new generation of U.S. Space Biologists". It can (i) serve as a nodal point for the synthesis of knowledge about gravitropism; (ii) provide a platform to space biology researchers to upload, visualize, and analyze Omics datasets of their choice in the context of plant metabolic and transport pathways, gene-networks, and plant developmental processes; (iii) extend the knowledge of genes and pathways that influence gravity response or intersect with other regulatory mechanisms (phototropism; chemotropism, and hormone signaling); and (iv) inform about evolutionary relevance of the conclusion derived from the Arabidopsis and rice (reference species in the Plant Reactome). Overall, our proposal support space biology researchers to formulate data-driven hypotheses for designing plants suited for cultivation in controlled environments in space flights, ISS, and beyond.
Task Progress:	The availability of a system-level in silico model of plant pathways serves as a framework to understand how various plant processes and overall plant development are affected or regulated by external stimuli, including gravity/microgravity. We identified 500 genes in Arabidopsis and rice that show differential expression in response to gravity. Furthermore, we improved gene functional annotations based on a critical review of published literature and analysis of transcriptome datasets available in public repositories, including NASA GeneLab (https://). Finally, using the Reactome Curator Tool, we manually curated in silico models of gravitropism pathways in rice and integrated these pathways into the Reactome Central Database. Using automated scripts, these reference rice pathways were used to derive the projection of pathways for an additional 129 species, ranging from unicellular photoautotrophs to higher plants, including several models and crop plants. The curated pathways and gene-orthology-based pathways are being made available on the Plant Reactome knowledgebase (https://). The work supported by this NASA grant --improved the content of the Plant Reactome, --two DiagramJs pathway widgets are embedded at https:// for pathway browsing and Analysis. Space biology researchers from this app can quickly connect to the Plant Reactome. Plant Reactome is an open-source, free-of-cost, publicly funded, plant pathway knowledgebase available to educators and researchers worldwide. Integrating the genes and pathways associated with gravitropism response in the Plant Reactome is expected to provide a platform to space biology researchers (i) in visualizing and analyzing their Omics datasets; (ii) in extending their knowledge of genes and pathways that influence gravity response or intersect with other regulatory mechanisms (phototropism; chemotropism, hydrotropism, and hormone signaling); and (iii) in formulating data-driven hypotheses for designing plants suited for cultivation in controlled environments in space. This proposal is well aligned with NASA's space biology goals, "Create new knowledge of how biological systems adapt to space that can translate into benefits for NASA Exploration Missions and for Earth applications through sponsoring competitively solicited Space Biology research" and "Train and inspire a new generation of U.S. Space Biologists".
Bibliography Type:	Description: (Last Updated: 02/12/2026)

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